

OmniGuard SOC

Autonomous Multi-Modal Threat Defense, Jitter-Invariant C2 Analytics & Graph-Based Incident Triage

AI/ML, SecOps & Cloud

IBM LinuxONE (s390x), AWS & On-Prem

PRIMARY DOMAINS

AI/ML, SecOps & Cloud

TARGET INFRASTRUCTURE

IBM LinuxONE, AWS & On-Prem

TARGET COMPLETION

December 2026 (12-14 Weeks)

THE TEAM

Khoi • Rahim • Aayush • Abu

Detection & Ingress Defense Leads

Phase 1 • Subsystems 1 & 2



Khoi

Encrypted C2 Behavioral Engine & Telemetry Pipeline

- ✔ **SpectraC2 Flow Pipeline:** Build flow-metadata engine on IBM LinuxONE (s390x) tracking IAT variance, packet symmetry, and spectral density.
- ✔ **FFT/PSD & LSTM Classifier:** Train and tune spectral feature extractors and PyTorch LSTM models to isolate jittered C2 beacons.
- ✔ **Telemetry Parsing:** Stand up Kafka/Redis streaming and Zeek/Suricata parsers feeding downstream graph models.

s390x LinuxONE

PyTorch LSTM

Zeek / Suricata



Rahim

Visual Ingress Defense & Adversary Emulation

- ✔ **VerifyEye Engine:** Develop YOLOv8/pHash brand-logo detection and Playwright DOM crawler for suspicious forms.
- ✔ **Endpoint Form Interception:** Build Chrome MV3 extension for real-time credential form locking on unverified submissions.
- ✔ **Adversary Emulation:** Script Flipper Zero BadUSB / rogue captive-portal scenarios and configure Sliver C2 listeners.

YOLOv8 + pHash

Chrome MV3

Sliver C2

Graph Correlation & Platform Leads

Phase 1 • Subsystems 3 & 4



Abu

Graph Correlation & Air-Gapped LLM Triage

- ✔ **Property-Graph Architecture:** Design Neo4j schema and entity resolution for hosts, users, IPs, and processes.
- ✔ **Blast-Radius Analytics:** Implement shortest-path, blast-radius, and centrality algorithms in Cypher / NetworkX.
- ✔ **Air-Gapped LLM Triage:** Deploy local Ollama (Llama-3-Cyber) triage engine and MITRE ATT&CK mapping prompts.

Neo4j Cypher

NetworkX

Ollama Llama-3



Aayush

Unified Dashboard & Platform Integration

- ✔ **Unified SOC Interface:** Build React / React Flow / D3.js dashboard: attack graphs, FFT heatmaps, timeline playback.
- ✔ **Real-time Streaming:** Wire WebSocket streaming for sub-second alert delivery across all three detection engines.
- ✔ **Cloud & Pipeline Orchestration:** Own AWS ECS/Lambda deployment and end-to-end integration across the five phases.

React / D3.js

FastAPI + Mongo

WebSockets / ECS

SpectraC2: Core Pipeline & AI Engine

Subsystem 1: Jitter-Invariant C2 Beacons Analytics • **Presenter: Khoi Duong**

Slide 1 / 2

PyTorch LSTM + Scikit-Learn

IBM LinuxONE s390x Ready

⚡ DUAL-DOMAIN FEATURE EXTRACTION ENGINE

- **14 Temporal Metrics:** Inter-Arrival Time (IAT) mean, variance, skewness, kurtosis, flow burstiness, and packet directional symmetry.
- **6 Spectral Metrics (FFT / PSD):** Fast Fourier Transform computing dominant frequency f_0 , Spectral Energy, PSD peak-to-average ratio, and Spectral Entropy.
- **Jitter Invariance:** Uncovers stealthy Cobalt Strike / Mythic sleep jitter algorithms where time-domain thresholding fails.

FFT Peak: $f_0 = 0.345$ Hz

Window: 50 packets

Sub-15ms Latency

🧠 AI ENSEMBLE & IBM LINUXONE READINESS

- **Sequential Classifier:** 2-layer PyTorch LSTM with Dropout (0.2) capturing non-linear inter-packet temporal dependencies.
- **Unsupervised Outlier Detector:** Isolation Forest (120 trees) detecting novel and zero-day beaconing topologies.
- **IBM LinuxONE s390x Native:** Multi-arch containerized build ([Dockerfile.s390x](#)) verified for enterprise zero-trust deployments.



bash: spectrac2-engine/inference/trainer.py

RUN: 1500 FLOWS

```
$ python spectrac2-engine/inference/trainer.py --samples 1500 --epochs 15

[*] Generating 1500 synthetic network flows...
[*] Training Isolation Forest model on temporal + spectral features...
[+] Isolation Forest saved to
spectrac2-engine/inference/models/isolation_forest.joblib
[*] Training PyTorch BeaconLSTM for 15 epochs...
    Epoch 5/15 | Loss: 0.0842 | Val Acc: 96.50%
    Epoch 10/15 | Loss: 0.0018 | Val Acc: 98.70%
    Epoch 15/15 | Loss: 0.0003 | Val Acc: 99.00%
[+] PyTorch BeaconLSTM saved to
spectrac2-engine/inference/models/spectrac2_lstm.pt
=====
[+] Precision: 98.10% | Recall: 100.00% | F1: 99.04%
=====
[*] Validation complete: Zero false negatives across jittered evaluation set.

✓ Model checkpoint exported to production staging directory.
```

98.1%
PRECISION

100%
RECALL

99.0%
F1-SCORE

Ingestion Pipeline & Live Detection

Subsystem 1 & 3: Streaming Telemetry, Automated Verification & Graph Handoff •

Presenter: Khoi Duong

Slide 2 / 2

27 / 27 Tests Passing
(100%)

Kafka + FastAPI + Graph
REST

🔌 NETWORK TELEMETRY PARSERS & STREAMING BRIDGE

- **Dual-Sensor Telemetry Support:** High-throughput streaming parsers for Zeek (`conn.log`, `ssl.log` in TSV/JSON) and Suricata (`eve.json` Flow & TLS).
- **Sliding-Window Aggregator:** Real-time 5-tuple flow grouping with resilient in-memory queue fallback for offline & CI testing.
- **FastAPI Microservice:** High-performance inference endpoints (`/classify-flow` and `/spectral`) with Pydantic v2 payload validation.

🔌 PHASE 2 HANDOFF & TEST SUITE INTEGRITY

- **Graph-Triage Linkage:** Consumer converts beacon detections into typed JSON alerts dispatched directly to the Graph-Triage Core REST API (`POST /api/v1/alerts`).
- **Robust Test Suite:** Comprehensive PyTest test harness spanning unit math, API endpoints, serialization, and stream aggregation.

27 / 27
PYTESTS PASSING

100%
PASS RATE (5.4S)

4.66s
JITTER TOLERATED

```
bash: pytest tests/ -v 27 PASSED

$ pytest tests/ -v
collecting ... collected 27 items
tests/test_feature_extraction.py::test_periodic_frequency PASSED [ 18%]
tests/test_inference.py::TestPredictor::test_predictor_e2e_beacon PASSED [ 48%]
tests/test_service_api.py::test_classify_flow_endpoint PASSED [ 55%]
tests/test_stream_pipeline.py::test_consumer_kafka_queue PASSED [ 74%]
tests/test_telemetry_parsers.py::TestSuricataParser PASSED [100%]

===== 27 passed, 1 warning in 5.43s =====
```

```
bash: stream_consumer.py --dry-run LIVE ALERT

$ python telemetry-pipeline/kafka-consumers/stream_consumer.py --dry-run
2026-09-18 23:56:12,004 [INFO] [SpectraC2-Consumer] [*] Running dry-run validation
with synthetic C2 beacon flow...

🚨 [C2 BEACON DETECTED] Host: WORKSTATION-CORP-42 | Severity: CRITICAL
Target: api-telemetry-cdn.net:443 (SSL/TLS)
Beacon Interval: ~2.9s | Jitter Observed: 4.6588s | Confidence: 64.3%
Action: Dispatched typed JSON payload to Graph-Triage REST endpoint (/api/v1/alerts)
```

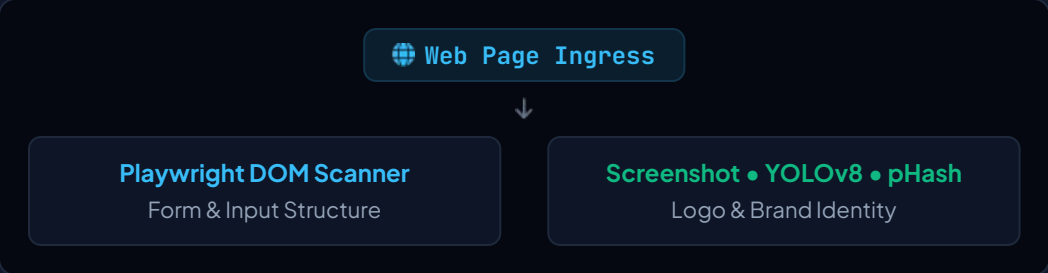
VerifyEye: Visual Ingress Defense & DOM Scanner

Subsystem 2: Multi-Modal Phishing Identification • Presenter: Rahim

Playwright + YOLOv8 + pHash

<500ms / Page

VerifyEye analyzes a webpage in multiple ways: it inspects the website's DOM structure looking for credential forms, login fields, and destination endpoints, coupled with computer vision brand detection.



HOW IT WORKS

Scans every on a page for two independent signals:

- ✓ Password field: `input[type="password"]`
- ✓ Username-like field: `input[type="text"]`, `input[type="email"]`, or name containing user / email

Flags `credential_form = True` only when both signals appear in the same form. Pure DOM inspection — <500ms per page.

```
bash: verifyeye-cli --url https://www.sfbu.edu/ ZERO FALSE POSITIVE

Enter website URL: https://www.sfbu.edu/

===== VerifyEye Result =====
URL: https://www.sfbu.edu/
Title: San Francisco Bay University | SFBU Fremont CA
Forms: 1
Credential Form: False
Username Field: True
Password Field: False
```

Live Validation — sfbu.edu
The one form on SFBU's homepage is a header search box, not a login form. VerifyEye correctly identified a text input without flagging a false credential-harvesting form.

Known Edge Case & Regression Testing
Any plain text input trips `username_field` (search boxes, names). Does not trigger false positives on `credential_form` since pairing with a password field is strictly enforced. Documented and regression-tested.

FastAPI, React Dashboard & MongoDB Ingestion

Subsystem 4: Alert Aggregation & SOC Presentation • **Presenter: Aayush**

Zero-Cloud-Egress

Pydantic v2 Schema

Backend — Alert API

- FastAPI service: GET /health, POST /alerts, GET /alerts
- Shared alert contract: 12 fields validated via Pydantic
- Invalid input rejected: bad severity returns HTTP 422
- Swagger docs at /docs self-documenting for team
- 4 fields map directly to Neo4j node types

Frontend — React Dashboard

- Vite + React dashboard fetching live from the Alert API
- Displays alert count and dynamic cards per alert
- Tracks: event type, severity, host, user, domain, status
- CORS scoped strictly to dashboard origin

Database — MongoDB

- MongoDB running locally; alerts persist across restarts
- Database omniguard, collection alerts
- Mongo holds raw alert; Neo4j stores graph topology
- Local, not cloud — maintains zero-cloud-egress posture

●●● mongodb: omniguard.alerts.findOne() • Sample Stored Document VALIDATED BSON

```
{ "_id": ObjectId('6aadcfdfe544d4566b4932b8'), "source":
"spectrac2", "event_type": "c2_beacon_detected", "severity":
"critical", "host": "WORKSTATION-CORP-42", "user":
null, "src_ip": "10.0.4.15", "domain":
"api-telemetry-cdn.net", "mitre_technique": "T1071", "status":
"new", "alert_id": "8e9755f6-cbda-418b-89c6-1447209f1651",
"description": "Encrypted C2 beacon loop detected on host WORKSTATION-CORP-42 targeting...", "timestamp":
ISODate('2026-09-18T23:57:19.393+00:00')} }
```

API Documentation & Integration Endpoints

Live Swagger Execution • Shared Data Contract Across Detection Subsystems

HTTP 200 OK

HTTP 201 CREATED

```
curl -X 'GET' 'http://127.0.0.1:8000/alerts' 200 OK

[ { "source": "verifyeye", "event_type":
"phishing_page_detected", "severity":
"critical", "host": "PERSIST-TEST", "domain":
"evil.com", "status": "new", "alert_id":
"a3639aa1-6322-4553-a3c6-91ca154eb931" }, {
"source": "spectrac2", "event_type":
"c2_beacon_detected", "severity": "critical",
"host": "WORKSTATION-CORP-42", "src_ip":
"10.0.4.15" } ]
```

```
curl -X 'POST' 'http://127.0.0.1:8000/alerts' 201 CREATED

{ "source": "verifyeye", "event_type":
"phishing_page_detected", "severity": "high", "host":
"MACOS-10", "user": "maharjan", "src_ip":
"10.0.0.42", "dst_ip": "198.51.100.42", "confidence":
0.633, "domain": "evil-microsoft-login.com",
"mitre_technique": "T1566", "status": "new",
"description": "Credential harvesting page
impersonating Microsoft 365 login", "alert_id":
"12b5c85b-2972-4b37-8c17-50a73b0876d8", "timestamp":
"2026-09-19T03:23:33.806808Z" }
```

Neo4j Graph Correlation & Blast Radius Analysis

Subsystem 3: Attack Path Visualization • Presenter: Abu

Cypher Query Engine

Blast Radius: 4 Hosts

NEO4J GRAPH ARCHITECTURE

- Built an OmniGuard test graph using **Users, Hosts, Processes, Domains, and IPs**.
- Connected security entities with relationships such as **CONNECTED_TO**, **SPAWNED**, **VISITED**, and **AUTHENTICATED_TO**.
- **Goal:** Identify attack route, lateral movement, and blast impact in real time.



SHORTEST PATH

2 Hops

WORKSTATION-01 → FILE-SERVER-01
→ DOMAIN-CONTROLLER-01

BLAST RADIUS

4 Hosts

Reachable internal hosts
compromised via lateral movement

Abu — Neo4j Implementation & Cypher Execution

Graph Schema Definition, Automated Entity Linkage & Lateral Path Exploration

Created 7 nodes • 6 relationships

Query Time: 547 ms

neo4j\$ MATCH (n)-[r]→(m) RETURN n, r, m; CYPHER 25

```
// Create entities and security relationships

CREATE (alice:User {name: 'Alice'}), (workstation:Host {name:
'WORKSTATION-01'}), (fileserver:Host {name: 'FILE-SERVER-01'}), (dc:Host {name:
'DOMAIN-CONTROLLER-01'}), (ps:Process {name: 'powershell.exe'}), (domain:Domain
{name: 'fake-microsoft-login.com'}), (ip:IP {name: '203.0.113.50'}) CREATE
(alice)-[:AUTHENTICATED_TO]→(workstation), (workstation)-[:VISITED]→(domain),
(workstation)-[:SPAWNED]→(ps), (workstation)-[:CONNECTED_TO]→(fileserver),
(fileserver)-[:CONNECTED_TO]→(dc);

✓ Created 7 nodes, created 6 relationships, set 7 properties, added 7 labels.

CREATE (alice:User {name: 'Alice'}), (workstation:Host {name:
'WORKSTATION-01'}), (fileserver:Host {name: 'FILE-SERVER-01'}), (dc:Host {name:
'DOMAIN-CONTROLLER-01'}), (ps:Process {name: 'powershell.exe'}), (domain:Domain
{name: 'fake-microsoft-login.com'}), (ip:IP {name: '203.0.113.50'}) CREATE
(alice)-[:AUTHENTICATED_TO]→(workstation), (workstation)-[:VISITED]→(domain),
(workstation)-[:SPAWNED]→(ps), (workstation)-[:CONNECTED_TO]→(fileserver),
(fileserver)-[:CONNECTED_TO]→(dc);
```

NODES (9) • RELATIONSHIPS (8)

Host (5) User (1) Process (1) Domain (1) IP (1)

CONNECTED_TO (5) SPAWNED (1) VISITED (1)

Query Results: shortestPath & blastRadius

Metric	Result	Details
shortestPath	2 hops	WORKSTATION-01 → FILE-SERVER-01 → DOMAIN-CONTROLLER-01
blastRadius	4 hosts	["FILE-SERVER-01", "DOMAIN-CONTROLLER-01", "HR-SERVER-01", "BACKUP-SERVER-01"]

SCHEDULE

PROJECT TIMELINE & KEY MILESTONES

5-phase agile schedule — 12–14 weeks

PHASE 1
Weeks 1–3

✓ DONE

1

PHASE 2
Weeks 4–6

📌 CP 1 (WEEK 5)

2

PHASE 3
Weeks 7–9

📌 CP 2 (WEEK 7)

3

PHASE 4
Weeks 10–12
Planned

4

PHASE 5
Weeks 13–14
Showcase

5

✓ COMPLETED (W1–W3)

INFRASTRUCTURE & DATA COLLECTION

Set up the LinuxONE/AWS infrastructure, Kafka and Redis, network telemetry, and testing environment

EXPECTED RESULT:

Working foundation for collecting security data

📌 ACTIVE: CHECKPOINT 1 (W5)

DETECTION & AI DEVELOPMENT

Develop the phishing/visual detection system, DOM analyzer, C2 analytics, and initial attack-graph structure

EXPECTED RESULT:

Initial threat-detection components (VerifyEye baseline, SpectraC2 FFT spectral features, Kafka streaming)

📌 NEXT TARGET: CHECKPOINT 2 (W7)

ATTACK CORRELATION & AI TRIAGE

Connect security events into an attack graph and develop the local AI triage system

EXPECTED RESULT:

Ability to connect events and summarize incidents (Neo4j multi-entity traversal, Dijkstra blast radius, local Ollama triage)

WEEKS 10–12

DASHBOARD & BROWSER PROTECTION

Build the SOC dashboard, Chrome extension, visualizations, and real-time alert streaming

EXPECTED RESULT:

Usable interface for security analysts (React Flow attack canvas, FFT heatmaps, 1-click quarantine)

WEEKS 13–14

TESTING & FINAL DEMONSTRATION

Run end-to-end attack scenarios, test the complete system, measure performance, and prepare the final demonstration

EXPECTED RESULT:

Complete OmniGuard SOC demonstration with >95% precision benchmark on APT intrusion replay

Phase 2 Timeline — Checkpoint 2 (Weeks 5–7)

Tracks: SpectraC2 • VerifyEye • Graph & UI — synchronized to feed Phase 3 (Ollama LLM Triage)

Checkpoints 4–6

Week	Khoi — SpectraC2	Rahim — VerifyEye	Aayush + Abu — Graph & UI
4	Wire <code>stream_consumer.py</code> → FastAPI predictor for live classification	Add cross-origin <code>form-action</code> validation; start logo dataset collection	Stand up Neo4j via <code>docker-compose.yml</code> ; define Cypher schema; sync on alert payload format
5	Deploy inference container on LinuxONE via <code>Dockerfile.s390x</code> ; validate latency	Train & quantize <code>YOLOv8</code> , export to ONNX, wire into <code>detector.py</code>	Build entity-resolution ingestion (<code>api/main.py</code>) against agreed schema
6	Benchmark LSTM / Isolation Forest vs. jitter flows; hit >95% precision	Favicon pHash comparison; end-to-end validation with trained model	React Flow / D3 graph canvas; WebSocket streaming; multi-pane layout

Week 6 = integration checkpoint — all three pipelines feeding Neo4j, ready to hand off into Phase 3.

ON TRACK FOR MILESTONE 2

Academic & Technical References

10 Citations

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[2] Lin, D., et al. (IEEE S&P) — Phishpedia: a deep-learning phishing detection system for logo ID, layout verification & coordinate mapping.

[3] Al-Eroud, A. & Alsmadi, I. — Identifying malicious HTTPS C2 traffic using behavioral flow metrics. *J. Info. Security & Applications*, vol. 72, 2023.

[4] Noel, S. & Jajodia, S. (IEEE TDSC) — Managing attack-graph complexity through visual hierarchies; foundation for blast-radius visualization.

[5] DARPA Transparent Computing Program — Graph-based provenance tracking & APT detection via fine-grained system telemetry, 2020.

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[8] IEEE S&P Workshops — Cyber threat intelligence extraction using domain-adapted local LLMs without privacy leakage, 2024.

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[10] OISF & Zeek Project — Suricata EVE JSON and Zeek network telemetry protocol parsing standards, 2024–25.